

FORTIGATE NSE4

Lab 3 — High Availability

1. Objective

Deploy two FortiGate firewalls as an Active-Passive High Availability cluster, providing configuration synchronization, session survivability, and automatic failover without manual intervention.

► 1.1 Why Active-Passive

Reason	Detail
Predictability	Only one unit processes traffic at a time, so behavior under load is deterministic
Simpler diagnosis	A single forwarding path means a packet either traversed the active unit or it did not
Common deployment	The usual choice in enterprise and industrial networks
Suits OT	Deterministic failover matters more than aggregate throughput where a process is running

Active-Active trades that away. It distributes sessions across both units for higher inspection capacity, at the cost of more complex session handling and harder fault diagnosis.

► 1.2 Prerequisites the Cluster Will Not Form Without

- Heartbeat interfaces directly connected between the two units, on a dedicated link
- Identical firmware version on both units
- Identical interface configuration on both units
- The same group name and password on both units

A mismatch in any of these produces a cluster that never forms, and the GUI does not always name which one is wrong.

2. Topology

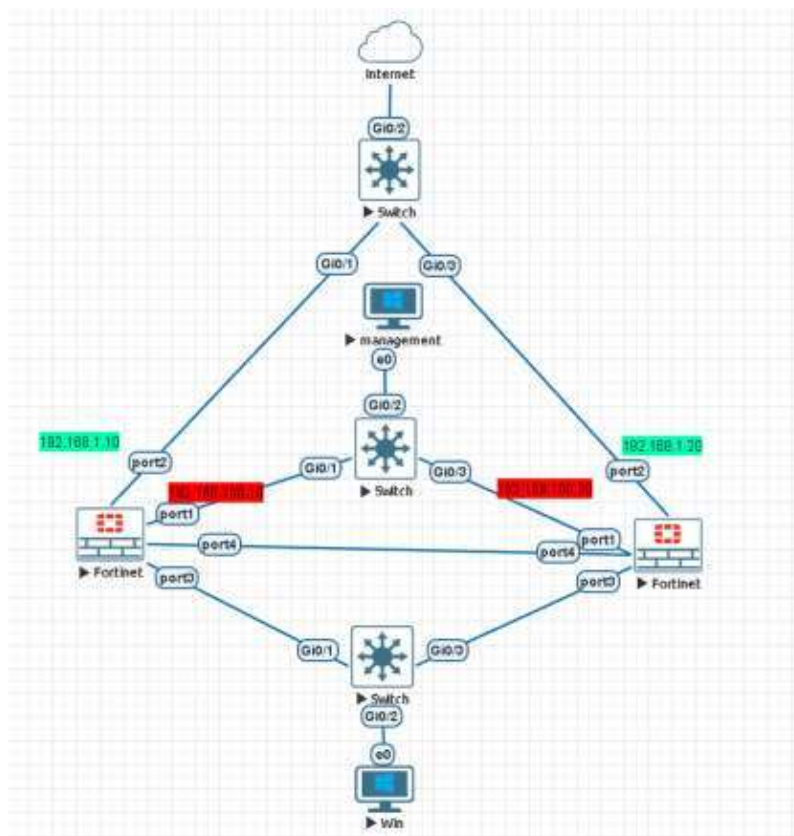


Figure 1 — Two FortiGates sharing LAN and WAN segments, joined by a dedicated heartbeat link on port4

Port	FortiGate 1 (Primary)	FortiGate 2 (Secondary)
port1 — MGMT	192.168.100.10/24	192.168.100.20/24
port2 — WAN	192.168.1.9/24(mistyped in the Screenshot)	192.168.1.20/24
port3 — LAN	192.168.20.1/24	synchronised from primary
port4 — Heartbeat	Direct link to peer	Direct link to peer

The heartbeat link is dedicated and direct. Carrying heartbeat traffic over a shared switch means a broadcast storm or a switch failure can sever it while both units remain healthy — producing split-brain, where both units believe they should be active.

3. Pre-HA Configuration

The primary is configured completely before HA is enabled. The secondary receives everything through synchronisation, so there is no need to duplicate the work by hand.

► 3.1 WAN Connectivity

```
execute ping 8.8.8.8
```

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default
       V - BGP VPNv4

Routing table for VRF=0
S*  0.0.0.0/0 [5/0] via 192.168.1.1, port2, [1/0]
C   192.168.1.0/24 is directly connected, port2
C   192.168.20.0/24 is directly connected, port3
C   192.168.100.0/24 is directly connected, port1
```

Figure 2 — WAN reachability confirmed before HA is introduced

► 3.2 Routing Table

```
get router info routing-table all
```

Physical Interface			
LAN (port3)	Physical Interface	192.168.20.1/255.255.255.0	
port1	Physical Interface	192.168.100.10/255.255.255.0	PING HTTPS SSH HTTP
port2	Physical Interface	192.168.1.9/255.255.255.0	
port4	Physical Interface	0.0.0.0/0.0.0.0	

Figure 3 — Default route via port2, with the connected subnets present

The default route points out of port2 toward the upstream gateway, and the three connected subnets are all present. This is the state that will be replicated to the secondary.

► 3.3 Interfaces

```
FortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=120 time=50.0 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=120 time=47.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=120 time=48.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=120 time=48.8 ms
```

Figure 4 — Interface list — port1 management, port2 WAN, port3 LAN, port4 heartbeat

port4 carries no address. A heartbeat interface does not need one — FGCP communicates over it directly, and assigning an IP would invite it to be used for something else.

► 3.4 Firewall Policy



Figure 5 — LAN to WAN policy with NAT enabled

A single outbound policy with NAT, permitting DNS, HTTPS and PING. This is enough to generate the continuous traffic the failover test depends on.

4. HA Configuration

► 4.1 Mode Selection

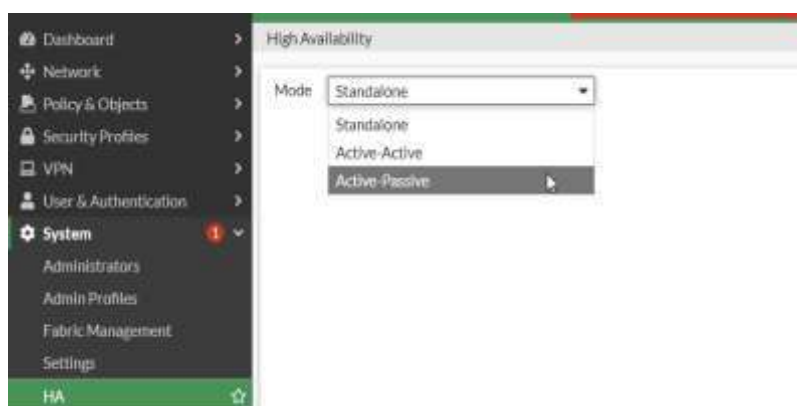


Figure 6 — System → HA — the default mode is Standalone

Mode	Behaviour
Standalone	No clustering. A single appliance and a single point of failure
Active-Active	Both units process traffic, with sessions distributed between them
Active-Passive	One unit forwards, the other stays synchronized and idle until failure

► 4.2 Primary Unit

Mode: Active-Passive
Device priority: 128

Cluster Settings

Group name: HA-PAIR
Password: *****
Session pickup: ☐

Monitor interfaces: port2, LAN (port3)

Heartbeat interfaces: port4

Figure 7 — Primary — priority 128, group HA-PAIR, heartbeat on port4

Setting	Primary	Purpose
Mode	Active-Passive	—
Device priority	128	Higher value wins the election
Group name	HA-PAIR	Must match on both units
Password	Shared secret	Must match on both units
Monitor interfaces	port2, port3	Interfaces whose failure should trigger failover
Heartbeat interfaces	port4	The dedicated peer link

Monitor interfaces are the mechanism that matters most here. Without them, HA reacts only to a unit failing entirely. **With port2 monitored, losing the WAN link alone is enough to hand the active role to the peer** — which is the scenario tested in section 6.

► 4.3 Secondary Unit

High Availability

Mode: Active-Passive
Device priority: 100

Cluster Settings

Group name: HA-PAIR
Password: *****
Session pickup: ☒

Monitor interfaces: port2, port3

Heartbeat interfaces: port4

Figure 8 — Secondary — priority 100, session pickup enabled

Setting	Secondary	Notes
Device priority	100	Lower than the primary, so it stays passive while the primary is healthy
Group name	HA-PAIR	Identical to the primary
Session pickup	Enabled	Replicates the session table so established connections survive a failover
Monitor interfaces	port2, port3	Matching the primary
Heartbeat interfaces	port4	Matching the primary

Session pickup is what separates a failover from an outage. Without it the secondary takes over with an empty session table, so every established connection is dropped and must be rebuilt. **A user would see the failover as a brief disconnection rather than as nothing at all.**

► 4.4 Primary Election

FGCP decides which unit becomes active by working down an ordered set of criteria.

Order	Criterion	Detail
1	Device priority	Range 0–255, default 128. Higher wins
2	Uptime	If priorities are equal, the unit running longer wins
3	Serial number	If both are equal, the higher serial number wins

After the cluster forms, the secondary loses independent management access — it is administered through the primary. This is expected behaviour and not a fault, but it surprises people the first time they try to reach the secondary's management address.

5. Verification

► 5.1 Cluster Status



Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
Synchronized	128	FortiGate1	FGVMEVH40SK04008	Primary	3m 46s	27	19.00 kbps
Synchronized	120	FortiGate-VM6	FGVMEVZF7FDQI0D5	Secondary	3m 48s	4	16.00 kbps

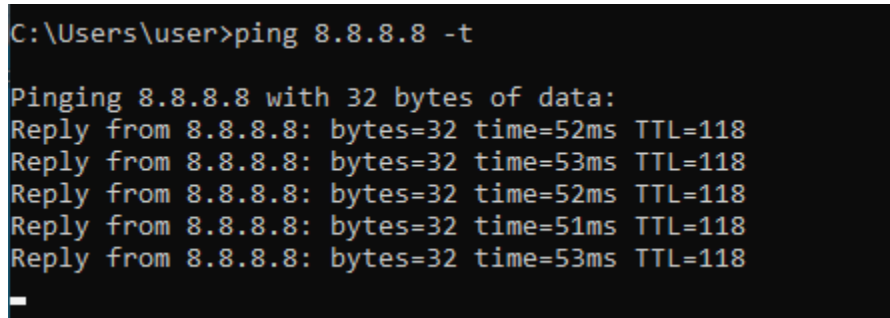
Figure 9 — Both units synchronised — primary handling sessions, secondary idle

Both units report **Synchronized**, the primary is carrying the session load, and the secondary sits at near-zero. Matching uptimes confirm both joined the cluster at the same time.

A note on the priorities shown. The dashboard reports the secondary at priority 120 rather than the 100 configured in section 4.3. **The election outcome is unaffected — the primary at 128 still wins — but the discrepancy is worth resolving before relying on the configuration**, since a value that does not match what was set suggests the change was not applied or was overwritten by synchronization.

► 5.2 Connectivity Baseline

```
ping 8.8.8.8 -t
```



```
C:\Users\user>ping 8.8.8.8 -t
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=53ms TTL=118
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=53ms TTL=118
-
```

Figure 10 — Continuous ping from the LAN client before any failure is induced

Consistent replies with stable round-trip times. This is the baseline against which the failover is measured — the ping must already be running before the failure, or the result understates performance.

6. Failover Testing

A cluster that has never been failed over is an assumption, not a control. Every HA deployment should be tested before it is relied upon, because a misconfigured heartbeat or an unmonitored interface produces a cluster that looks healthy and does nothing when it is needed.

► 6.1 Scenarios

Test	What it exercises
Power off the primary	Total unit failure — heartbeat stops entirely
Disconnect the WAN port	Partial failure — the unit is alive but has lost a monitored interface
Disconnect the heartbeat link	Peer visibility loss — tests the split-brain condition

The second scenario is the interesting one. A powered-off unit is easy to detect. **A unit that is running, responding on the heartbeat, and simply cannot reach the internet is only detected because port2 is a monitored interface** — without that, the cluster would keep the failed unit active.

▶ 6.2 Expected Behaviour

- The secondary assumes the active role within one to five seconds
- A continuous ping loses at most one or two packets
- Established sessions survive, because session pickup is enabled

▶ 6.3 WAN Link Removal

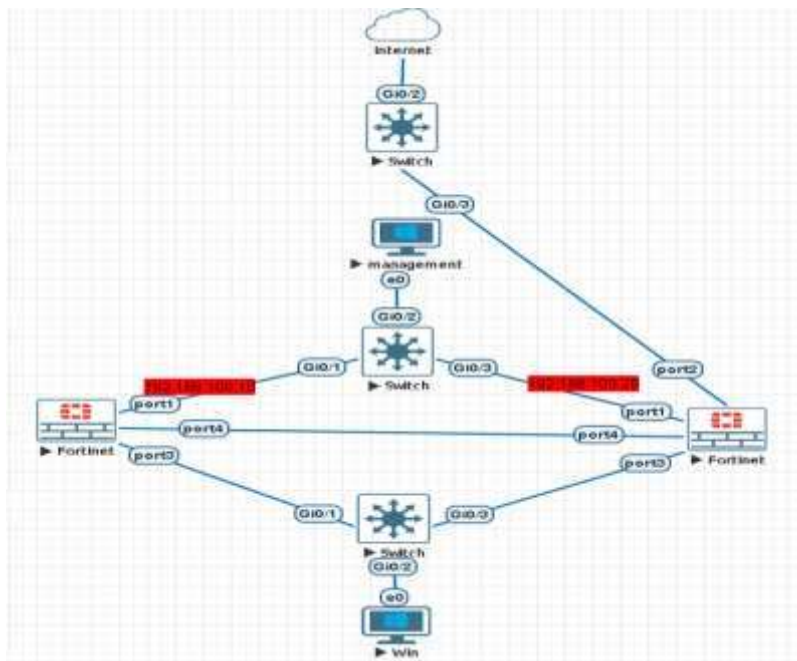


Figure 11 — The WAN link removed from the primary to simulate an upstream failure

```
Reply from 8.8.8.8: bytes=32 time=50ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=55ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=52ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
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Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=50ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
Reply from 8.8.8.8: bytes=32 time=51ms TTL=118
```

Figure 12 — Continuous ping across the failover — no interruption

The ping stream shows no timeouts and no gap in the sequence. **The secondary assumed the active role and continued forwarding without a single lost packet** — better than the one or two expected, because the interface failure was detected immediately rather than through missed heartbeats.

Why this is faster than a power-off failover: losing a monitored interface is an immediate, locally observable event. A powered-off peer must first be inferred from missed heartbeats, which takes as long as the heartbeat interval and threshold allow.

7. FGCP

FortiGate Clustering Protocol is the proprietary mechanism that builds and maintains the cluster. It runs over the heartbeat interfaces and handles everything the cluster does automatically.

► 7.1 What FGCP Handles

Function	Detail
Heartbeat monitoring	Detects peer failure over the dedicated heartbeat links
Configuration sync	Replicates the primary's configuration to the secondary continuously
Session sync	Replicates the active session table when session pickup is enabled
Primary election	Decides the active unit by priority, then uptime, then serial number
Virtual MAC	The cluster presents a shared MAC address, so a failover requires no ARP relearning by surrounding devices

The virtual MAC is why failover is transparent. Without it, every device on the segment would hold an ARP entry pointing at the failed unit's hardware address and would keep sending to it until that entry aged out — turning a sub-second failover into a multi-minute outage.

► 7.2 Compared with Cisco FTD Failover

Feature	FortiGate (FGCP)	Cisco FTD
Protocol	FGCP, proprietary	Cisco failover protocol
Heartbeat	Dedicated heartbeat interfaces	Failover link plus state link
Session sync	Session pickup, configurable	Stateful failover, always on
Role selection	Priority, higher wins	Primary/secondary role assignment
Config sync	Automatic from primary	Automatic from active
Virtual MAC	Yes	Yes

8. Summary

Component	Detail
HA mode	Active-Passive
Protocol	FGCP
Primary election	Priority 128 against 100; range 0–255, default 128
Heartbeat	port4 — dedicated, directly connected, no IP address
Monitor interfaces	port2 WAN and port3 LAN
Session pickup	Enabled — established connections survive the transition
Config sync	Automatic from primary; the secondary loses independent management access
Failover tested	WAN link removal from the primary
Measured result	Zero packet loss across the transition
Verification	Cluster status dashboard plus continuous ping from the LAN client